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This Historic Community Is Pushing the Nation Toward a Wind Power Revolution

Block Island, off the New England coast, overcame political strife to lead the way on energy independence



A vessel nears the commercial wind farm 3.8 miles off the coast of Block Island. Neil Ever Osborne

By Elizabeth Royte**Photographs by Neil Ever Osborne**

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Block Island, 15 miles off the coast at its farthest point, has always been at the mercy of the four winds. Raging winter gusts have been known to rip porches off houses and knock stones off the rock walls that lattice the island's meadows and pastures. More regularly, breezes delivered to residents the drone of enormous diesel-burning generators, the Rhode Island community's sole source of power. No one liked it, "but that was just part of island life," a local real estate agent tells me. People got used to the noise, and those who lived near the power plant—less than half a mile from downtown—resigned themselves to frequently scrubbing soot from their windows and sills.

But then, at precisely 5:30 a.m. on the first of May, 2017, a great silence fell upon the land. The generators, after roaring for 89 years, shut down. And yet electrons continued to flow.

"Suddenly you could hear the leaves rustling, the waves breaking, and the birds"—Henry duPont, a local engineer who attended the diesel shutdown, breaks off, allowing the twitter and squawk of spring migrants to speak in his stead. Residents have been marveling at the quietude ever since.

Since that day, Block Island has been the only community in the United States fully powered by offshore wind: in this case, five 6-megawatt turbines pounded into the seafloor just south of the island's Mohegan Bluffs. Over the next several years the Block Island venture will be joined by many more towns and cities, as up to 2,000 new turbines begin to populate utility-scale wind farms along the Atlantic Seaboard. These projects were fast-tracked a year ago when

President Biden set a national goal of generating 30 gigawatts of offshore wind energy on both coasts and in the Gulf of Mexico by 2030. That's enough juice to run ten million homes while avoiding the production of 78 million metric tons of greenhouse gas emissions.

“The loudest people against the wind farm considered the island their little paradise.”



The Mohegan Bluffs community faces the open Atlantic—and the array of towering wind turbines. Neil Ever Osborne

From its inception, the Block Island Wind Farm, launched by a Providence-based company called Deepwater Wind, was meant to be a demonstration project. Not of the technology—European nations nailed that decades ago and now operate more than 5,000 offshore turbines—but of the knotty permitting process that allows a commercially financed power generator to

plug into an established electrical grid. And smoothing the regulatory path will be essential if the nation is going to quit fossil fuels. According to a recent Princeton University study, total installed wind power must grow more than sixfold from today's capacity for the nation to achieve net-zero greenhouse gas emissions by mid-century.

While tens of thousands of smaller wind turbines, rated at less than two megawatts, have stippled the American landscape since the 1970s, truly widespread wind power will depend on much larger devices. Wind speed tends to increase with altitude, and the taller the tower is, the larger the blades and the turbine can be, dramatically increasing energy production. For example, one of the tallest turbines in operation, General Electric's Haliade-X, a 13-megawatt behemoth installed in Rotterdam, reaches about 80 stories high, and each blade is 351 feet long. In just seven seconds it generates enough power to serve the average American home for a day.

These giants will almost certainly be planted primarily at sea, where it's easier to transport enormous blades and tower sections, there's more space for arrays, and permitting hassles with property owners are reduced. "You're not dealing with people's backyards or other kinds of challenges over hundreds of miles," says Matt Morrissey, a former vice president of Orsted, the Danish company that acquired the Block Island Wind Farm in 2018.



Hundreds of feet above the sea, Will Fancher, a rope access supervisor for the Block Island Wind Farm, performs maintenance that is far from routine. Neil Ever Osborne



Fancher working on the turbine blade's tip receptor. Neil Ever Osborne



Fancher inspects the turbine blade's tip receptor. Neil Ever Osborne

Offshore wind, especially in the Northeast, is stronger and more consistent than terrestrial wind because the ocean surface creates less drag than land, which is pocked with trees, buildings and mountains. Steady ocean temperatures also make for steadier wind. And with companies planning to site wind farms 12 or more miles from shore, reducing the ability to see them from living room windows, they've quelled some of the opposition that doomed the nation's first proposed offshore wind farm, off Cape Cod, Massachusetts.

But not all of it.

On a damp May morning, when 80 percent of Block Island homes are still shuttered for the off-season, I meet with Dick Martin, who has been with the Block Island Power Company for more than four decades. Until the wind farm came online, Martin tended BIPCo's five giant yellow diesel generators, which he shows me in metal sheds lined with din-dampening

padding. Back in his office, he lets me heft a foot-long sample of the submarine cable that runs three miles from the wind farm to Block Island, and then another 20 miles to the mainland. As thick as a spiral ham, the cable section weighs 46 pounds.

Under heavy skies, we drive to Southeast Lighthouse, atop Mohegan Bluffs. From here, the first of the five wind towers, which form a nearly straight line, appears much closer than 2.9 miles. The turbine towers reach more than 330 feet above the sea surface. The white blades, each one 241 feet long, spin mesmerically, disappearing into the clouds. “I think they’re beautiful,” Martin says. Sculptural, others pronounce. Majestic.



Wind turbine at sunset, Block Island Wind Farm, Rhode Island. Neil Ever Osborne

I have no aesthetic problem with these symbols of high modernity. Would I feel different if I lived in one of the mansions on this bluff, my expansive ocean views interrupted by a power plant? Possibly. But altered views were not the argument that opponents most often voiced at public meetings held over several years before construction. “People said they were concerned

about the birds, the fish, the whales, the noise—anything they could throw at it,” Martin recalls. Others worried that drilling would further erode the bluffs (it didn’t), or that the blades would create mind-altering flicker effects (who knows?).

“The loudest people against the wind farm were those who considered the island their summer getaway, their little paradise,” says Kim Gaffett, a naturalist whose family has lived on Block Island for generations and who served as the head of the island’s town council during the project’s development. “They flew in from Washington State, New Jersey and Massachusetts to fight it.” Most of the roughly 1,000 year-round residents—who paid some of the highest electric bills in the country—favored the turbines.

Everyone wanted data on environmental impacts. And so two years before Deepwater dropped any equipment into the water, it hired environmental consultants to study the area’s fin- and shellfish, marine mammals and birds. Data from these studies and from abroad show that offshore wind farms don’t appear to harm migratory birds (most of which travel closer to shore). The impact on bats has not been sufficiently studied. High-definition and thermal-imaging cameras and other equipment placed on a vessel monitoring the work area indicated that marine mammals navigate away from wind farms during pile driving (which lasted a few days per turbine) and had no problem avoiding the piles thereafter.



Kim Gaffett, weighing a bird in her workshop, led the town council during the debate about installing offshore turbines. Neil Ever Osborne



One visitor to Block Island is a migratory bird called the northern parula, held by Kim Gaffett, a naturalist. Neil Ever Osborne

Fish studies that continued during construction and for three years afterward revealed that the Block Island Wind Farm did not destroy but actually created marine habitat: mussels and barnacles quickly colonized the underwater structures, attracting fish. Now the turbines are the place to be if you want to hook a black sea bass, tautog, fluke or bluefish. As for lobster, well, this never was a prime lobster congregation spot—one reason the farm was sited here—though some evidence suggests that the electromagnetic fields of underwater cables may affect lobster navigation.

Were the turbines a hazard to navigation, as some fretted? Lit, numbered and marked on charts, no. But the scads of boaters now drawn to them created another problem. As one charter captain told researchers, “When I fish, I prefer to fish alone ... And that area now, you can’t fish in that area and be alone.”

The National Audubon Society generally supports offshore wind, on the principle that climate change will do more harm to birds than wind farms will. So does the National Wildlife

Federation. “We’ve got to stand up large-scale sources of green energy” to face the “urgent threat” of climate change, the NWF’s Catherine Bowes says, adding, “offshore wind energy holds incredible potential.”

To protect whales—fewer than 350 North Atlantic right whales remain—scientists and environmentalists have come up with guidelines for offshore wind developers along the Atlantic coastline. They should alter construction schedules to accommodate whale migration; slow down vessels when whales are in the vicinity; and use noise-dampening technologies, like devices that create sound-muffling curtains of air bubbles around piles, to avoid injuring whales’ sensitive ears or driving them into areas of increased boat traffic. Another option is to use “gravity-based” turbine supports, which can weigh several thousand tons and sink onto, rather than being pounded into, the seafloor. Best practices also include surveying and pile driving only during daylight hours so work can stop if whales are spotted.

“Lessons from Europe don’t exist for large baleen whales” like humpbacks, blue and right whales.



Lisa Nolan, executive director of the Southeast Lighthouse Foundation, which received a donation of more than \$1 million from the wind farm's developer. Neil Ever Osborne



Island resident Steve Miller hooks a bluefish near a turbine tower. Turns out the parts of the structures that are underwater actually attract many forms of marine life. Neil Ever Osborne

There's a general feeling that wind farms and related port infrastructure should not be placed in habitats that are critical to whales. But what's critical? According to Howard Rosenbaum, who monitors whales for the Wildlife Conservation Society, researchers are still learning when, where and how these peripatetic and long-lived creatures occupy the New York Bight, a busy waterway that will likely see increased vessel traffic as offshore wind is developed. "Lessons from Europe don't exist for large baleen whales" like humpbacks, fin, blue and right whales, he says. Because most of the proposed energy sites on the Atlantic Coast lie within whale migration routes, he says, "the race is on to gather baseline data." Fortunately, he adds, "everyone—the states, the feds, developers, NGOs—are all very engaged on this issue."

Tooling along Block Island's hilly roads, past picturesque ponds and glades of white-flowering shadbush, I find it hard to ignore the signs of privilege: the muted elegance of the historic

homes and the abundance of open space given over to neither crops nor animals. More than 47 percent of island land is conserved, which makes Block Island, via Thoreau (“a man is rich in proportion to the number of things which he can afford to let alone”), rich indeed. Passing one weathered-gray shingled estate after another, I think there is at least some justice in the placement of those steel turbines in the backyards of the island’s wealthiest residents. After all, nationwide, it’s low-income communities that live closest to, and suffer the worst health impacts from, dirty power plants.

I roll down a private lane on the island’s southwest lobe to meet with one of the wind farm’s most persistent critics, David Lewis. The Lewis family has lived continuously on Block Island since at least 1817, and today their property is set on a rise surrounded by 200 acres of stone-walled pastures, another 200 acres of publicly owned conservation land, and the Atlantic Ocean.

“This was not a NIMBY issue for me,” Lewis says, gesturing toward the wind-whipped sea. It’s true that you can’t see all five turbines in a single visual gulp from this spot, but moving between the upstairs windows, you can glimpse each one. A Harvard-trained biologist and a recently retired manufacturing representative for General Dynamics, Lewis objected to the wind farm because of the political process that, he said, led to the power purchase agreement with National Grid, the utility company. Now all Rhode Islanders were paying too much for clean energy, he and others contend, while wind farm developers made a killing. Yet a recent analysis of Block Island power rates, which vary by season, found that the cost per kilowatt hour, if averaged over a year, is 44 percent lower than it was before the wind farm went online.



The towers are spaced about half a mile apart, weigh 440 tons apiece and are anchored to the seabed in water about 80 to 90 feet deep. Neil Ever Osborne

Lewis presses a loftier point, even now that the issue is moot. “Seascapes are sacred and immutable,” he tells me, then quotes from *Moby-Dick*, when the *Pequod* sinks beneath the waves “and the great shroud of the sea rolled on as it rolled five thousand years ago.” At a public hearing before Rhode Island’s Coastal Resources Management Council, he testified against his white whale. “One of the irreplaceable features of many outward-looking seascapes is that...one sees an expansive view to an open horizon that has looked just the same for longer than man can contemplate....Only the arrogance of man allows him to choose a point in time to say, ‘Here and now, I have the right to permanently alter the way something has always been, into something else of my choosing.’”

Yes, it’s nice when pretty things stay that way, I feel like saying. But coastlines have always changed; this headland didn’t always host the Lewis family compound, for example. More important, how is society supposed to manage a trade-off in which a pleasant view enjoyed by

a handful of people is weighed against the benefits of a sustainable energy supply that would accrue to many thousands of people, now and into the future?

Five years on, the wind farm dust has mostly settled. Block Island fishermen are happy, charter boats run turbine tours for intrigued tourists, the fiber-optic strand Deepwater included in the submarine power cable has started to provide the island with fast internet, and property values have held.



A section of the submarine power cable linking wind farm, island and mainland. Note the thin fiber optic cable, too. Neil Ever Osborne

“Change is difficult for everyone, but new people coming in accept the wind farm,” says Cindy Pappas, of Sullivan Sotheby’s International Realty. Of course, she adds, “No one is saying ‘I want a windmill view.’”

Most people acknowledge the power they receive now is better than it used to be. Energy from the diesel generators sputtered, often blowing out appliances, and brownouts in the summertime, when the population triples, were common. Now folks are buying air conditioners, a perfect example of what is known as the Jevons paradox, in which efficiencies (such as an endless supply of carbon-free energy at a stable price) spur greater consumption.

For better or worse, the northern Atlantic is now acknowledged to be an almost textbook place for offshore wind development. The continental shelf is wide and relatively shallow (in deeper California waters, and in Maine, developers are planning for floating turbines anchored to the seafloor with cables); the region has plenty of nearby customers, which reduces transmission costs; and the wind is, as they say in Rhode Island, wicked good. “The Gulf Stream runs up the coast unimpeded, as does the wind,” Orsted’s Morrissey says. “By the time it reaches southern New England, it’s at its apex.”

“We are an entirely green community. We did something to make our children proud.”



Equipped with lights and noted on charts, the Block Island wind turbines don't pose an unusual danger to mariners, Henry duPont says: "You could sail through them blind." Neil Ever Osborne

The East Coast also offers what's called peak coincidence. "Offshore wind reaches its greatest speed in late afternoon, when the turbines are really spinning," Morrissey says. "And that's exactly when the large power centers in Boston and Providence and New York and Washington require the most energy."

Lessons from Block Island now inform plans for wind farms up and down the East Coast. Developers, residents, businesses, planners, regulators and scientists have learned to work with each other. Other island communities have noted the additional benefits the community extracted from Deepwater Wind, such as the fiber optic cable to the mainland and two donations of some \$1.25 million each to the local historical society and the Southeast Lighthouse Foundation.

But there is still much to sort out as the wind industry marches toward the Biden administration's goals.

The nation's energy transition will require major investments in new ports, work boats, manufacturing facilities and grid upgrades. Utilities still need sustainable and far cheaper ways to store energy for wind farms' inevitable lulls in generation. Developers need certainty that states will buy their product. The cost of offshore power must continue to go down, as it has done on land; the price of *onshore* wind power has dropped from 7 cents to 2 cents per kilowatt hour in the last decade. And commercial fishers need assurance that they will be compensated financially if they lose access to fishing grounds.

"We need to keep cumulative [wildlife] impacts in mind if we are going to be doing this from Maine to Virginia," Sue Tuxbury, a fisheries biologist with the National Oceanic and Atmospheric Administration, told an offshore wind-energy science forum hosted by the University of Rhode Island's department of oceanography. Birds, fish and whales are unlikely to suffer much harm from the Block Island wind farm, but no one knows how a thousand turbines might affect those creatures, especially floating turbines, whose cables could potentially entangle large whales. There are, to date, few floating arrays or even models to study.



The higher you go, the more wind there is. Ocean breezes in the Northeast tend to be reliable, creating a prime target for wind farm developers. Neil Ever Osborne

Engineers are tweaking wind farm designs to increase efficiency. For 20 years, Princeton University engineer Elie Bou-Zeid says, turbines have been getting bigger, but “wind farm output can increase by up to 60 percent through smarter layouts.” Bou-Zeid is studying designs that cluster turbines in groups of three; others are studying arrays in which the blades of front-row turbines are tilted to produce less power so the second and third rows can produce even more.

Block Islanders, for the most part, remain righteously proud of the part they played in laying the groundwork for what promises to be a massive buildout of offshore wind on the East Coast. “Even though we are the smallest town in the smallest state in the nation, we’re taking an enormous step forward not only for ourselves, but for the country and ultimately the planet,” says Bryan Wilson, manager of the Block Island Wind Farm. “We learned so much

about how to do this right, and now we are an entirely green community. We did something to make our children proud.”

And for those who rue the industrialization of the seaboard? One is tempted toward pragmatism: Things change. New Englanders no longer hunt whales to produce commercial oils. Block Islanders no longer burn peat to heat their homes, as they did after deforesting their island in the early 1700s. The peat bogs they excavated are now those lovely ponds.

Through a contentious but deliberative process, Block Islanders severed themselves from a power plant that burned a million gallons of fossil fuel a year. That’s not a vast amount in the overall scheme of things, but it’s a model that countless people are aching to follow.

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